Sahaj Bhadja

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Summary

Graduate software engineer with 2+ years building **full-stack product features**, **database-driven workflows**, and **tested cloud-ready applications** across requirements, development, QA, deployment, and iterative release cycles.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Delivered **full lifecycle backend features** from requirement analysis through deployment, building FastAPI workflows for 5K+ monthly jobs with validation, metadata consistency checks, and production-focused release controls.
- Improved database reporting and retrieval performance for 500+ users by optimizing schema/query patterns, reducing latency from **1.2s to 350ms** and increasing consistency of user-facing analytical outputs.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Built **React and Flask application modules** handling 10K+ daily requests, implementing secure file workflows, user-facing interfaces, and structured API responses across end-to-end grading and feedback product flows.
- Strengthened software quality with automated testing and CI/CD checks, reducing release turnaround by **60%** while maintaining stable behavior across iterative deployments in a rapidly evolving product environment.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Developed and maintained **application and data integration services** with AWS Glue, S3, and Python/Java components, improving response times by **30%** and enabling reliable enterprise data management workflows.
- Owned test automation and versioned release support with Jenkins, PyTest, and Mockito, improving CI efficiency by **68%** and reducing technical debt by **20%** across collaborative Agile delivery cycles.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Created Python pipelines for 100K+ timeseries-like trajectories, producing reproducible analysis outputs and technical documentation that improved cross-team decision-making on data quality, model behavior, and implementation priorities.
- Built debugging diagnostics and reporting artifacts to accelerate issue isolation, enabling faster turnaround from experimental findings to actionable engineering tasks within multidisciplinary scientific and software teams.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Implemented REST/WebSocket features supporting 2M+ synchronized records and eight workflows, improving data consistency and operational responsiveness in application interfaces used by distributed stakeholders.
- Automated deployment, file-handling checks, and repeatable test workflows with Docker and Jenkins, reducing release time by **50%** while improving traceability of software versioning and delivery status.

Skills

Application Development: Python, Java, JavaScript/TypeScript, Flask, FastAPI, React
Data & Reporting: SQL, PostgreSQL, MySQL, database reporting, data collection and analysis, metadata-aware workflows
Testing & Release: unit/integration testing, CI/CD, version control (Git), Jenkins, GitHub Actions, automated validation
Engineering Practices: requirements modeling, documentation, file/document management, Agile/Scrum collaboration

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Built a full-stack fintech product supporting 50K+ users and 1M+ transactions with secure workflows, data-rich interfaces, and reliable deployment practices aligned to high-availability financial application standards.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Developed a schema-aware SQL workspace for 10+ databases with validated query generation, categorized outputs, and structured retrieval flows that improved analysis speed and reduced manual troubleshooting effort.

Linguistix | *Python, FastAPI, AWS ECS/Fargate, CodePipeline*

- Implemented a containerized API with automated quality checks, versioned deployment pipelines, and cloud runtime controls, achieving **87%** processing accuracy while strengthening release repeatability and production monitoring readiness.